

Research Statement

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As a behavioral economist, I combine game theory and laboratory experiments to document and explain how boundedly rational behavior shapes competition and coordination. I study how people acquire and process information and reason about one another under different information structures and strategic environments. My goal is to use this understanding to guide the design of market mechanisms and communication rules.

1. Consumer Search and Position Prominence

My job market paper, “Auction for Prominence and Price-Directed Search,” studies how competition for search prominence, the advantage of being inspected first, affects prices and consumer search, and how these effects vary with the cost of inspecting a second product. In online marketplaces, basic product information, including prices, is readily available, but finding a good match remains costly. Because consumers rarely inspect every option, sellers pay for search ads and sponsored listings to draw consumers’ attention to their products.

I develop a model in which prices are observable before search, allowing sellers to influence consumers’ search decisions through their pricing. Sellers’ willingness to pay for prominence is determined endogenously by the resulting difference in expected revenues between positions. In the model, two sellers bid for the prominent position in a second-price auction and then set prices. The buyer observes both prices and learns for free whether the prominent seller’s product is a good match. She then decides whether to incur a search cost to inspect the other seller’s product. I test the model’s predictions for bids, prices, and buyers’ search and purchase decisions in a laboratory experiment using low- and high-search-cost treatments.

The experiment confirms that costlier search strengthens the pricing advantage of prominence, while revealing systematic departures from the equilibrium benchmark. At both search costs, prices are more dispersed than the mixed-strategy equilibrium predicts, while the average price gap is smaller than predicted. These price patterns alter buyers’ search incentives. Buyers generally follow the predicted search cutoffs, but both under-search and over-search errors are more common after a bad match with the prominent seller, when finding a suitable alternative matters most. Average bids rise with the search cost, but less than predicted, and exceed the realized difference in revenues between positions at both cost levels. These findings suggest that the design of markets for prominence must account for behavior both in the pricing and search that determine what prominence is worth, and in the bidding process that determines who obtains it.

I plan to extend this research in two directions related to the design of markets for prominence. First, I will compare auctioned prominence with random position assignment, keeping the pricing and search stages unchanged. Comparing welfare across these mechanisms will help assess whether a market mechanism is an effective way to allocate prominence. Second, I will extend the model by allowing a revenue-seeking platform to choose the search cost. I will test experimentally whether and how the platform benefits from this control, including by exploiting potential behavioral biases suggested by my current results. This work will examine the welfare consequences of platform control over search frictions and inform platform regulation more broadly.

2. Depth of Strategic Reasoning

My work on bounded rationality examines where people start when reasoning about others and how far they proceed. Understanding these limits helps explain players' initial responses to unfamiliar or complex decision environments. In joint work, "Revisiting the Intra-Team Communication Method to Elicit Level-k Reasoning in Beauty Contests and 11–20 Games," we provide the first replication of Burchardi and Penczynski's beauty-contest experiment and extend their intra-team communication method to variants of the 11–20 game. Participants explain their choices in incentivized messages to teammates, allowing us to infer their depth of reasoning and compare non-strategic players' behavior with more sophisticated players' beliefs about it.

In the beauty contest, elicited beliefs about non-strategic play are consistent with the observed behavior of players classified as non-strategic, as in the original study. However, this belief-action consistency breaks down in the reordered 11–20 game: non-strategic choices are more dispersed than higher-level players expect, and both choices and beliefs depart from the standard assumption. Moreover, using our within-subject design, we find that individuals' reasoning levels inferred from messages show positive correlations across games, a relationship not revealed by choices alone. This provides tentative support for viewing depth of reasoning as a stable individual characteristic across strategic settings.

3. Communication and Coordination

In a third project, I study how the rules of communication affect coordination when players face asymmetric information about the state and uncertain signal delivery. In "Coordination under Uncertainty and Noisy Communication: A Generalized Email Game," one player knows whether the state favors a risky joint action, while the other player does not. In the favorable state, a single signal can reveal the state to the uninformed player, but fails to arrive with a fixed probability. The model permits both an inefficient equilibrium in which the signal is ignored and a more efficient one in which players act on it, making equilibrium selection the central empirical question.

In the laboratory, I vary the communication protocol to examine how voluntary sending and its cost affect coordination. I find that relative to a protocol that sends the signal automatically, free voluntary sending by the informed player makes players more likely to take the risky action after favorable information. Adding a sending cost, however, deters informed players from sending and reduces overall action on favorable information, despite a slight increase in recipients' responsiveness to delivered signals. These results suggest that choosing to send can convey an intention to coordinate when players' interests are aligned, even when uncertain signal delivery prevents the state from becoming common knowledge. A fee intended to enhance signal credibility can nevertheless backfire by discouraging communication.

4. Research Pipeline

I plan to bring my research on how people reason and communicate into the design of market mechanisms. I will extend my search model to allow multiple sellers to compete for positions with different search costs. Consumers will choose where to search next and when to stop. By explicitly modeling consumers' search paths, I aim to connect the literatures on ordered search and sponsored-search auctions. I will compare auction formats in the laboratory to identify which delivers the highest efficiency and assess welfare distribution among consumers, sellers, and platforms. These comparisons will inform platform design and regulation.